

Implementing the Fama-French five-factor model for the German stock market

Philipp Dirkx*
Zeppelin University Friedrichshafen

Franziska J. Peter†
Zeppelin University Friedrichshafen

December 13, 2018

Abstract

We implement the Fama-French five-factor model for the German market using recent monthly data from 2002 to 2017. We construct the five factors associated with the market, size, value, profitability, and investment for the CDAX constituents and examine to which extent the five-factor model captures the return premia in the German market. The results show that in comparison with the three factor model, the five factor model does not add significant explanatory power to the analysis. We conclude that, the validity of the profitability and investment factors within the context of international asset pricing studies, cannot be transferred to the country specific case of the German market.

Keywords: asset pricing; Fama-French five factor model; German stock market; CDAX

*PhD student, Department of Empirical Finance and Econometrics, Zeppelin University, Am Seemossener Horn 20, 88045 Friedrichshafen, Germany.

†Department of Empirical Finance and Econometrics, Zeppelin University, Am Seemossener Horn 20, 88045 Friedrichshafen, Germany. Phone: 0049 7541 6009 2231. Mail: franziska.peter@zu.de.

1 Introduction

Recently, Fama and French (2015) introduced a five-factor asset pricing model which augments their three-factor model (Fama and French (1993)) by the profitability and investment factors. While Fama and French (2015) focus on the U.S. market, Fama and French (2017) extend the analysis to a global reach, covering North America, Europe and Asia Pacific. Country-specific studies, however, are rare, despite the fact that results in Fama and French (2012), Fama and French (2017), and Griffin (2002) suggest that the global version of the factor asset pricing model is not overall convincing, and local versions might provide more insights with respect to explaining anomalies. Among previous country specific studies are those by Daniel et al. (2001), who examine asset pricing anomalies in the Japanese Stock market and L'Her et al. (2004), who apply the Carhart (1997) four-factor model to the Canadian stock market. The Fama-French five-factor model is tested on the Japanese market by Kubota and Takehara (2018) and for Australian data by Chiah et al. (2016).

To the best of our knowledge the five-factor model has not been implemented for the German stock market yet. We fill this gap by using recent monthly data of the CDAX constituents from 2002 to 2017 and analyze whether and to which extent the Fama-French five-factor model offers superior explanatory power over the Fama-French three-factor model.

The German market has been previously studied by Ziegler et al. (2007), who apply the Fama-French three-factor model to the German equity market and cover a time period from 1968 to 1995. They also extend the model with two bond-linked factors to analyze the fit. They conclude, that the explainability of the three-factor model in Germany is not as high as in the U.S., however, it constitutes an improvement compared to a one-factor model based on the Capital Asset Pricing Model. Schmidt et al. (2017) show within the context of a broad international study, that the German equity market had a significant negative size premium between 1991 and 2012. Glaser and Weber (2003) analyse in detail the momentum effect in the German market, and Hanauer et al. (2011) apply the Carhart

four-factor model to the German equity market relying on a sample covering the period 1996 until 2011. They summarize, that the four-factor model only yields a slight improvement over the three-factor model and that the factors for the German market show little correlation to other international studies. Artmann et al. (2012) conduct an analysis on the German equity market, covering the one-, three- and four-factor model. They analyze the period from 1960 until 2006 and construct portfolios based on different methods for sorting. They conclude, that the results are sensitive to the construction of the portfolios and point out, that the four-factor model delivered the best results in terms of explanatory power. The paper proceeds as follows: Section 2 introduces the data, Section 3 outlines the factor construction as well as preliminary results on the constructed factors, Section 4 applies the three- and five-factor model to the German stock market and Section 5 concludes.

2 Data and methods

While researchers have identified numerous factors over the last 50 years (e.g. Banz, 1981 and Harvey et al., 2016) Fama and French justify the identification of their additional two factors by relying on the dividend discount model (c.p. Miller and Modigliani, 1961) to derive the profitability and investment factor. Based on the dividend discount model the present value of a stock, m_t , at time t is given by the discounted, expected dividend

$$m_t = \sum_{\tau=1}^{\infty} \frac{E(d_{t+\tau})}{(1+r)^\tau} \quad (1)$$

where $E(d_{t+\tau})$ denotes the expected dividend per share for period $t + \tau$ and r denotes the long-term expected internal rate of return on dividends. Fama and French (2015) slightly adjust Equation (1) to show the relationship between the expected return, expected

profitability, expected investment and the book-to-market ratio. In particular they show that the total market value, M_t , of a stock at time t is equal to

$$M_t = \sum_{\tau=1}^{\infty} \frac{E(Y_{t+\tau} - dB_{t+\tau})}{(1+r)^\tau} \quad (2)$$

where $Y_{t+\tau}$ denotes total equity earnings for period $t + \tau$ and $dB_{t+\tau}$ gives the delta of the book value between $t + \tau$ and $t + \tau - 1$. Subsequently, they divide the whole term by the book value in time t , B_t

$$\frac{M_t}{B_t} = \frac{\sum_{\tau=1}^{\infty} E(Y_{t+\tau} - dB_{t+\tau}) / (1+r)^\tau}{B_t}. \quad (3)$$

Equation (3) implies that, ceteris paribus, a) a higher book-to-market ratio implies a higher expected return; b) higher expected earnings (profitability) imply a higher expected return; c) the higher the growth in book equity (investment), the lower the expected return. Based on these results, Fama and French (2015) add two additional factors to the Fama-French three-factor models and propose to estimate the following five-factor model:

$$R_{it} = a_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + \varepsilon_{ti} \quad (4)$$

where R_{it} denotes the return of portfolio i at time t . $RMRF_t$ gives the excess return of the market return over the risk free rate at time t . SMB_t relates to the size factor, based the difference between the return of a portfolio of small stocks minus the return of a portfolio consisting of large stocks. The value factor is captured by HML_t , where the return difference between a portfolio with a high book-to- market (B/M) ratio and a portfolio with a low B/M ratio is taken. The first newly introduced factor, RMW_t , captures the potential profitability premium and is calculated as the return difference of two portfolios referring to high profitability and low profitability companies. The second newly introduced *investment*

factor, CMA_t , measures the difference between the return of a portfolio of stocks with low investment and the return of a portfolio consisting of stocks with high investment.

In order to implement the model outlined in Equation (4), we rely on monthly returns of the CDAX companies, covering a time period of 15 years, ranging from June 2002 and June 2017. The data are from the Bloomberg database. Previous asset pricing studies by Ziegler et al. (2007), Hanauer et al. (2011), and Artmann et al. (2012) have excluded financials from the investment universe, reasoning that the market values financial companies differently and different accounting rules apply. Fama and French (2015), on the other hand, do not exclude financials. Within our analysis we focus on the results including financials, however, we also conduct the same set of analyses for a sample excluding financials and, if applicable, outline any differences. The CDAX includes all Xetra-listed companies of the prime and general standard. The portfolios are recalculated in the end of June of every year. Only companies, which have a positive book equity value are used. The total return comprises the price return of an equity and the reinvested dividend. When applying all filters necessary to calculate the SMB, HML, RMW and CMA factors, the average number of analyzed stocks including financials is 370 and excluding financials 325, while the number of stocks never drops below 294 and 275, respectively.

The smaller amount of available entities compared to the U.S. market, leads to the adjustment of calculating 16 (4x4) value-weighted portfolios rather than 25 (5x5) as in Fama and French (2015). This corresponds to previous studies of the German equity market by Artmann et al. (2012) and Hanauer et al. (2011). The correlation of the returns of the adjusted investment universe with the CDAX returns amounts to 0.98. Consequently, we assume that our sample is representative for the overall German equity market. The newly constructed basket of remaining companies will serve as the market portfolio for the regression calculations. The risk-free rate will be based on the one month EURIBOR rates.

3 Factor construction and preliminary analyses

The risk factors SMB, HML, RMW and CMA are calculated as defined by Fama and French (2015) and can be described as independent 2x3 sorts. All factors are based on two size groups and three groups for the according other sorting variable. The size breakpoint of all factors is the median market capitalization of the sample companies. Concerning the valuation (B/M), operating profitability (OP), and Investment (Inv) factors, the breakpoints are determined by the 30th and 70th quantiles of the sample companies.

The operating profitability is calculated using all accounting numbers from the end of the previous fiscal year. It is defined by the annual revenues minus the cost of goods sold, interest expenses, selling-, general- and administrative expenses, divided by the book equity. Investment is defined as the ratio of total assets of a stock at the fiscal year end of $t - 1$ and the total assets at fiscal year end of $t - 2$.

Independent sorts are used to assign stocks into two size and three groups according to B/M, OP, and Inv groups. As for the size groups, there is a classification into small (S) and big (B). Considering valuation (B/M), stocks are classified as high (H), neutral (N) and low (L). As for the operating profitability (OP), stocks are classified as robust (R), neutral (N) and weak (W). Stocks are grouped into three investment groups (Inv), being conservative (C), neutral (N) and aggressive (A).

The size factor (SMB) is calculated by three size factors $SMB_{B/M}$, SMB_{OP} and SMB_{Inv} given by

$$\begin{aligned} SMB_{B/M} &= (S_H + S_N + S_L)/3 - (B_H + B_N + B_L)/3 \\ SMB_{OP} &= (S_R + S_N + S_W)/3 - (B_R + B_N + B_W)/3 \\ SMB_{Inv} &= (S_C + S_N + S_A)/3 - (B_C + B_N + B_A)/3 \end{aligned}$$

The final factors SMB, HLM, RMW, and CMA are then derived as

$$SMB = (SMB_{B/M} + SMB_{OP} + SMB_{Inv})/3$$

$$HML = (S_H + B_H)/2 - (S_L + B_L)/2$$

$$RMW = (S_R + B_R)/2 - (S_W + B_W)/2$$

$$CMA = (S_C + B_C)/2 - (S_A + B_A)/2$$

Table 1 shows three times 16 value-weighted excess returns (average return over risk-free rate) for the Size-B/M, the Size-OP and the Size-Inv portfolios. The value-weighted portfolios are based on independent sorts of the corresponding two variables. The row always refers to the size quartile, while the column refers to the valuation (B/M), the profitability (OP) or the investment (Inv) quartile.

Table 1: Average monthly returns, the standard deviation and the t-statistic of the 4x4 value-weighted portfolios including financials for June 2002 until June 2017.

Average returns					Standard deviation					T-statistic				
Size-B/M	Low	2	3	High	Size-B/M	Low	2	3	High	Size-B/M	Low	2	3	High
Small	0.45	0.63	0.94	0.64	Small	6.60	4.99	4.33	5.24	Small	1.15	2.02	3.29	1.96
2	1.04	0.68	1.06	0.74	2	6.21	4.45	5.15	5.57	2	2.51	2.41	3.07	2.08
3	0.82	0.85	0.83	0.83	3	4.81	4.62	4.93	6.33	3	2.63	2.83	2.59	2.01
Big	0.77	0.75	1.36	1.05	Big	5.71	5.24	6.02	6.62	Big	2.10	2.25	3.31	2.39
Size-OP	Low	2	3	High	Size-OP	Low	2	3	High	Size-OP	Low	2	3	High
Small	0.57	0.51	1.12	0.58	Small	5.48	4.77	4.76	6.56	Small	1.70	1.79	3.50	1.44
2	1.20	1.02	0.87	0.65	2	5.36	4.43	5.11	5.17	2	3.30	3.46	2.59	2.00
3	0.80	0.95	1.23	1.01	3	6.03	4.93	5.21	5.22	3	2.05	2.93	3.47	2.91
Big	0.42	0.43	0.73	0.77	Big	6.36	5.14	5.18	5.84	Big	1.15	1.46	2.21	2.06
Size-Inv	Low	2	3	High	Size-Inv	Low	2	3	High	Size-Inv	Low	2	3	High
Small	0.71	0.72	0.85	0.65	Small	5.59	5.56	5.47	5.39	Small	2.00	2.04	2.39	1.91
2	0.89	0.88	1.07	0.82	2	5.35	4.02	4.70	5.95	2	2.54	3.37	3.39	2.13
3	1.73	1.07	1.04	0.92	3	6.33	4.64	4.60	6.25	3	3.93	3.43	3.38	2.24
Big	0.07	0.49	1.06	0.58	Big	6.42	5.00	5.31	6.09	Big	0.40	1.65	2.99	1.55

For small and big stocks, the average excess return of the portfolios tend to increase with rising book-to-market ratio. Opposing international research, the returns for almost all portfolios increase with market capitalization. The relationship between B/M, size and excess returns of the German market confirms the findings of Schrimpf et al. (2007), Hanauer et al. (2011), and Artmann et al. (2012). Within the international study of Fama and French (2017), it is also detected, that the size effect in Europe for the Size-B/M portfolios is not existent and only weak in North America for the period of 1990 to 2015. Considering the Size-OP portfolios in the second Panel of Table 1, we do not observe a clear pattern of increasing average returns with size. However, the two highest size quartiles show increasing average excess returns with increasing operating profitability. This partly corresponds to the results of Fama and French (2015) who report their Size-OP portfolios of U.S. stocks to show a size effect across all OP quintiles, while excess returns are steadily increasing with higher operating profitability. In their international study Fama and French (2017) find a consistent increase in returns for higher operating profitability across all size quintiles for Europe. With respect to the size-investment sorted portfolios, panel 3 in Table 1 reveals no notable change in average excess returns with increasing investment. For the two large size quartiles, the spreads increase sharply, however with different directions of change. There is no clear relationship between increasing or decreasing investment and average excess returns. Fama and French (2015) find a size effect in four out of five Size-Inv quintiles and decreasing returns with increasing investment across all size quintiles. Their described relationship between investment and returns for the U.S. is similar to the returns in Fama and French (2017) for Europe, however, not as pronounced.

Considering the constructed factors, Table 2 provides an overview of the descriptive statistics of the five factors as well as their correlations. The average return for the market factor RMRF is 0.74%, while being strongly statistically significant. The SMB factor yields on average a return of -0.06%, while showing no statistical significance. This is coherent with other studies on the German market, however opposing Banz (1981) and results from in-

Table 2: Descriptive statistics and correlations of the constructed factors.
R (market) denotes the market return constructed from the sample stocks, R (risk free) denotes the risk free rate.

Variable	Mean	St.Dev.	T-Test
R (market)	0.86	5.31	2.18
R (risk free)	0.12	0.12	13.01
RMRF	0.74	5.33	1.87
SMB	-0.06	3.56	-0.22
HML	0.43	2.90	1.98
RMW	0.11	2.78	0.54
CMA	0.12	2.69	0.61

Correlations					
	RMRF	SMB	HMLS	RMW	CMA
RMRF	1				
SMB	-0.62	1			
HML	0.28	-0.25	1		
RMW	0.21	-0.26	0.06	1	
CMA	-0.31	0.35	0.16	-0.3	1

ternational asset pricing studies, which claim that small capitalized companies outperform large capitalized companies. Covering the German market, Schrimpf et al. (2007) calculated a SMB factor average return of -0.19% per month between 1969 and 2002. This result is almost identical in the study of Artmann et al. (2012), with an average SMB return of -0.18% for the period between 1962 until 2006. Interestingly, Ziegler et al. (2007) covered a slightly shorter period from 1968 until 1995 and calculated an average return for the SMB factor of 0.08%. In contrast, Hanauer et al. (2011) covering the period from 1996 until 2011 find a strongly negative SMB average return of -0.71%. This leads to the observation that small companies seem to have performed particularly bad in comparison to large companies in the end of the 1990s and the 2000s. In an international context, The SMB factor for Fama and French (2015) between 1963 and 2012 is significantly positive with an average return of 0.29% and a test statistic of 2.24. However in their international study (2017), none of the regions shows a statistically significant positive or negative SMB factor. The HML factor calculation has the highest average return of the long/short calculated factors,

amounting to 0.43%. This result is coherent with German and international studies in terms of the average return and its statistical significance. The RMW factor has an average return of 0.11%, while not being statistically significant. Fama and French (2015, 2017) report statistically significant RMW factors with average returns of 0.26 to 0.40% for the U.S. and Europe, while Japan and Asia Pacific have average RMW returns of 0.12 and 0.20% respectively. The average return for the CMA factor only yields 0.12% with a high standard deviation and low statistical significance. In comparison, only Japan had weaker average returns for the CMA factor with 0.09% per month, while the U.S., Europe and Asia Pacific have average returns between 0.22% and 0.39% (Fama and French, 2017).

In terms of correlation it is worth to highlight, that the market factor RMRF is strongly negatively correlation to SMB (-0.617) and the CMA factor (-0.309). The SMB factor on the other hand is negatively correlated to HML (-0.253) and the RMW factor (-0.264). The strongest positive correlation exists between the average returns of SMB and CMA (0,349). For the U.S., Fama and French (2015) find that the RMRF factor is only positively correlated to the SMB factor, while being negatively correlated to all other factors, which is contrary to the results for Germany. While the SMB factor is actually also negatively correlated to the HML factor in their study, there is no positive correlation between the SMB and CMA factor. The difference in observation can probably in part be attributed to the observed inverse size effect in the German market, compared to the U.S. market.

Figure 1 shows a graphical depiction of the development of the factors over time. The market factor (RMRF) factor exhibits the highest cumulated performance, while showing the highest variation. The HML factor performed strongly until the end of 2011, continuously generating negative returns afterwards. On the contrary, the SMB factor underperformed until the end of 2014 and continuously gained in value afterwards. The RMW factor underperformed until the beginning/middle of 2007, basically shortly before the start of the financial crisis and then recovered strongly afterwards. The CMA factor shows a continuous gain until the end of 2011 and drifts downward afterwards. While we will make continuous

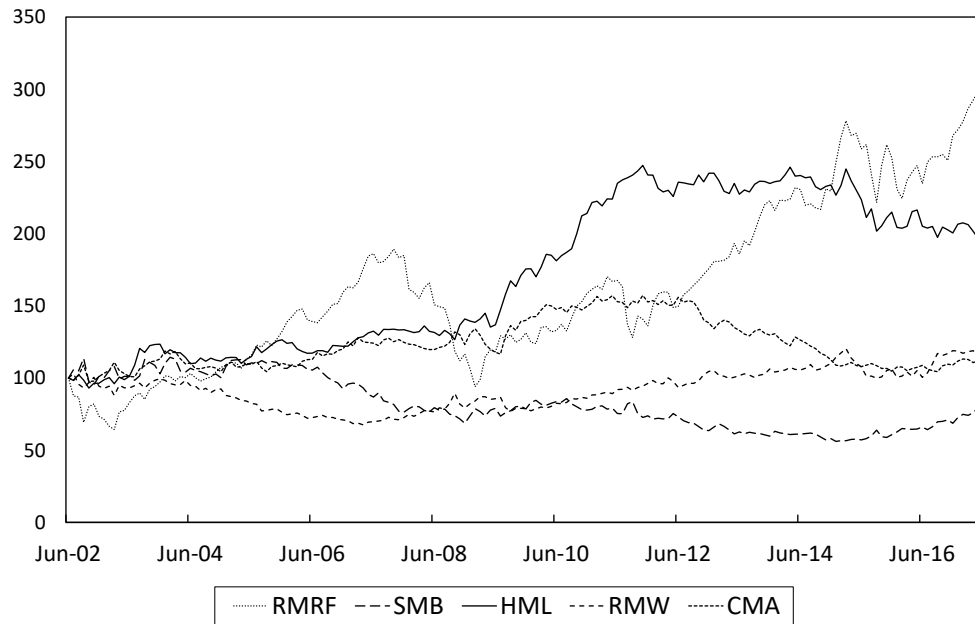


Figure 1: Development of the constructed factors over time.

comparisons of results to other studies covering the German equity market, the comparisons need to be taken with a grain of salt. Brueckner et al. (2015) reveal, that results for the same analysis period can differ based on the data provider, mainly due to data quality. Hanauer et al. (2011) also offer a depiction of the development of the factors, ranging from July 1996 until December 2011. The development of their SMB and HML factor match the factor performance in Figure 1 pretty well, accounting for the different time horizons. Their HML factor continuously increases between the beginning of the 2000s until the end of their study in 2011 and their SMB factor stagnates between the beginning and middle of the 2000s and then underperforms after that until the end of 2011.

4 Empirical results of the three- and five-factor model for the German equity market

Multi-variate regressions are performed to examine the relationship between returns of stock portfolios and stock related characteristics. The returns to be explained are the three sets of 16 value-weighted portfolios, namely Size-B/M, Size-OP and Size-Inv. The stock related characteristics are expressed through the factors, the explanatory variables. To start off, a three factor model regression is performed to explain the returns of the 16 Size-B/M portfolios with the market, size and value factor. To compare those results and the fit of the model, the five factor model regression is performed to explain the excess returns of each of the 16 value-weighted Size-B/M, Size-OP and Size-Inv portfolios.

Table 3 gives an overview of the results for the three factor model regressions given by

$$R_{it} = a_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + \epsilon_{ti} \quad (5)$$

Considering the coefficients on the market factor, b_i which range from 0.82 to 1.14, no clear relationship with increasing or decreasing size or B/M ratio is observable, except that the largest estimates are observed for the high B/M group. The coefficients for s_i decrease with increasing size. This observation is stable throughout all B/M groups, while the coefficients even become negative for the biggest size and largest two B/M quartiles. The t-statistics indicate statistical significance for the three smallest size quartiles, while no significant effect is detected for the biggest size quartile. The coefficients for h_i are positive and significant different from zero with strong test statistics for the highest B/M group. However, the other B/M quartiles show little factor loading for HML. While size does not have any significant impact on the h_i coefficients, one can observe a tendency for rising coefficients with rising B/M ratio within all size quartiles. Considering the intercepts, α , we observe only two statistically significant intercepts at a 10% significance level. The adjusted R^2 detects the

Table 3: Results for the Fama-French three-factor model.

Regression estimates (left panel) for the different portfolios are based on equation (5) together with the corresponding t-statistics (right panel).

Estimates for α					$t(\alpha)$				
Size/BM	Low	2	3	High	Size/BM	Low	2	3	High
Low	-0.28	0.32	0.22	-0.14	Low	-0.70	0.94	0.85	-0.56
2	0.09	0.07	0.12	-0.12	2	0.47	0.35	0.77	-0.66
3	0.38	0.36	0.12	0.49	3	1.19	1.78	0.60	1.86
High	0.07	-0.04	0.02	0.04	High	0.59	-0.30	0.14	0.18
Estimates for b					$t(b)$				
Low	0.94	1.03	0.82	1.00	Low	10.09	12.61	13.60	16.65
2	1.03	0.90	0.97	1.05	2	22.31	20.58	26.37	23.81
3	0.84	1.03	0.97	1.14	3	18.32	21.31	21.31	18.38
High	0.98	1.02	1.05	0.83	High	32.76	30.47	26.14	14.97
Estimates for s					$t(s)$				
Low	1.19	0.96	0.93	1.09	Low	8.54	7.88	10.42	12.15
2	1.03	0.96	1.04	1.14	2	15.00	14.88	19.10	17.49
3	0.59	0.64	0.58	0.79	3	8.68	8.91	8.50	8.60
High	0.03	0.11	-0.15	-0.06	High	0.65	2.25	-2.47	-0.75
Estimates for h					$t(h)$				
Low	0.21	0.01	0.12	0.54	Low	1.53	0.10	1.37	6.05
2	-0.40	0.00	0.17	0.39	2	-5.73	0.05	3.10	5.91
3	-0.07	-0.09	0.05	0.18	3	-0.92	-1.18	0.80	1.96
High	-0.37	0.07	0.04	0.92	High	-8.24	1.45	0.66	11.12
R^2									
Low	0.38	0.46	0.52	0.66					
2	0.74	0.71	0.80	0.78					
3	0.65	0.73	0.73	0.67					
High	0.90	0.89	0.88	0.78					

worst fit for 0,70 on average and most stable for the biggest the smallest size quartile. Our results partly confirm the findings by Ziegler et al. (2007), whose analysis covers the period from 1968 until 1995, for the more recent data of our sample period. However, we do not find the increasing importance of the HML factor with rising B/M value as reported by Ziegler et al. (2007).

Subsequently, we augment the three-factor model by the two additional factors referring to investment and profitability and estimate the model given in equation (4). The results of the 16 Size-B/M portfolios are presented in Table 4. Adding the factors profitability and investment to the three factor model does not lead to a drastic change in results in terms of coefficients and test statistics for the 16 Size-B/M portfolios. The coefficients the market factor, SMB, HLM, change only marginally for the five factor model regression, compared to the three factor model regression for the Size-B/M portfolios. Considering RMRF, the strongest coefficients can be found in the highest B/M quartile, while the s_i estimates are strongest in the smallest size quartile, stable through all B/M quartiles, and continuously decrease with increasing size of the stock portfolios. Considering HML, a decent tendency of increasing coefficients with increasing B/M ratio is observable, just like for the three-factor-model on the same LHS portfolios.

In the lowest B/M quartile, three out of four coefficients are negative while the highest B/M quartile shows three strongly positive and statistically significant coefficients. The newly added profitability factor leads to the surprising result, that most coefficients on the market factor are negative, while only the subset of the highest two B/M quartiles and the biggest size quartile have strong positive coefficients. We do not find any relationship between the profitability factor loading and increasing or decreasing size or valuation. The results are similar for the second, newly introduced factor, the investment factor. Only the highest B/M quartile shows four positive coefficients.

The results of the five factor model and the size-OP portfolios are given in Table 5. The coefficients on the market factor, b_i , for the Size-OP portfolios in Table 5 are statistically significant positive and show no relationship between increasing or decreasing size or operating profitability quartile. The size coefficients indicate a very similar result to the Size-B/M portfolios, stable across the various OP quartiles and continuously decreasing with bigger size quartiles. Again, the test statistics are large and strongly positive for all of the first three size quartile portfolios. The factor loadings for HML are very diverse with no clear relationship. The coefficients for the profitability factor are not affected by size in an strong and homogeneous way. As for increasing operating profitability, one can find three strongly negative coefficients in the lowest OP quartile, while the coefficients continuously increase with increasing operating profitability. In the highest OP quartile, three coefficients are positive and two of those show strong test statistics of 5.67 and 10.31. As already observed for the Size-B/M portfolios, the CMA factor loadings are very small and mostly not significantly different from zero. On average, only a slight improvement in adjusted R^2 can be observed. The intercepts are rather small in general, however three out of four intercepts for the lowest and the highest OP quartile are negative, while all eight portfolios for the middle two OP quartiles show positive intercepts. Four intercepts show statistical significance at 5%, however, there is no clear relationship in terms of size or profitability observable.

Table 6 shows the results of the five factor model and the size-OP portfolios. The coefficients on the market factor for the Size-Inv portfolios are rather volatile and range from 0.77 to 1.19. The SMB factor loading gives the same impression as for the Size-B/M and Size-OP portfolios, with rather stable coefficients through all investment quartiles and decreasing coefficients with increasing size. The coefficients on HML show no relationship with increasing or decreasing size or investment. Most test statistics mirror no statistical significance. The coefficients for profitability are surprisingly all negative, except for the portfolios in the second highest investment quartile and in the biggest size quartile. The

coefficients on the investment factor show a clear relationship, being positive with high test statistics between 2.03 and 6.57 in the lowest investment quartile, decreasing steadily with increasing investment to negative coefficients. The intercepts show the highest range of values and t-statistics of all performed regressions, ranging from -1.88 to 3.07 in terms of t-statistics, while 5 intercepts are statistically significant at a 5% level.

We also conduct Gibbons, Ross, Shanken (GRS) tests according to Gibbons et al. (1989), testing the joint significance of the intercepts (a_i) in equation (4) for the three double sorts (Size/BM, Size/OP, and Size/Inv). The corresponding p-values are given by 0.632 for Size/BM, by 0.019 for Size/OP and by 0.006 for Size/Inv double sorted portfolios. Consequently, we find joint significance of the pricing errors at the 1% significance level for the Size/Inv sorted portfolios, only, while we find evidence that the Fama-French five-factor model captures the cross-sectional return variation when considering Size/BM sorted portfolios and to a lesser extent for Size/Op sorted portfolios.

Table 4: Results for the Fama-French five-factor model for Size-Valuation double sorted portfolios.

Regression estimates (left panel) for the different portfolios are based on equation (4) together with the corresponding t-statistics (right panel).

Estimates for α					$t(\alpha)$				
Size/BM	Low	2	3	High	Size/BM	Low	2	3	High
Low	-0.27	0.36	0.24	-0.13	Low	-0.67	1.06	0.99	-0.51
2	0.10	0.07	0.13	-0.11	2	0.49	0.40	0.84	-0.60
3	0.41	0.39	0.14	0.50	3	2.23	1.99	0.76	2.02
High	0.10	-0.03	-0.02	0.02	High	0.85	-0.21	-0.11	0.08
Estimates for β					$t(\beta)$				
Low	0.93	1.03	0.83	1.02	Low	9.79	12.61	13.95	17.32
2	1.05	0.90	0.97	1.06	2	22.53	20.17	25.83	24.63
3	0.84	1.04	0.98	1.17	3	18.92	21.95	21.70	19.49
High	0.97	1.04	1.06	0.83	High	33.18	32.94	27.48	14.84
Estimates for s					$t(s)$				
Low	1.20	0.92	0.88	1.01	Low	8.27	7.39	9.71	11.25
2	0.99	0.96	1.04	1.09	2	13.96	14.17	18.33	16.60
3	0.57	0.60	0.55	0.70	3	8.35	8.30	8.00	7.64
High	0.04	0.05	-0.15	-0.05	High	0.96	1.13	-2.53	-0.57
Estimates for h					$t(h)$				
Low	0.24	0.03	0.12	0.51	Low	1.61	0.26	1.30	5.62
2	-0.43	0.01	0.18	0.37	2	-5.91	0.14	3.17	5.54
3	-0.04	-0.08	0.07	0.13	3	-0.61	-1.10	0.98	1.45
High	-0.33	0.05	-0.01	0.90	High	-7.19	1.04	-0.19	10.44
Estimates for r					$t(r)$				
Low	-0.03	-0.34	-0.32	-0.31	Low	-0.19	-2.62	-3.42	-3.39
2	-0.14	-0.07	-0.07	-0.24	2	-1.93	-0.97	-1.18	-3.57
3	-0.31	-0.30	-0.26	-0.36	3	-4.46	-4.00	-3.68	-3.87
High	-0.14	-0.25	0.26	0.18	High	-3.15	-5.14	4.27	2.01
Estimates for c					$t(c)$				
Low	-0.09	-0.10	-0.01	0.09	Low	-0.54	-0.70	-0.07	0.89
2	0.09	-0.03	-0.05	0.05	2	1.12	-0.37	-0.81	0.71
3	-0.12	-0.04	-0.07	0.15	3	-1.52	-0.45	-0.86	1.39
High	-0.18	0.06	0.21	0.08	High	-3.41	1.14	3.04	0.81
Adjusted R^2									
Low	0.39	0.49	0.55	0.69					
2	0.75	0.71	0.81	0.80					
3	0.69	0.75	0.76	0.71					
High	0.91	0.91	0.89	0.79					

Table 5: Results for the Fama-French five-factor model for Size-Operating Profitability double sorted portfolios.

Regression estimates (left panel) for the different portfolios are based on equation (4) together with the corresponding t-statistics (right panel).

Estimates for α					$t(\alpha)$				
Size/OP	Low	2	3	High	Size/OP	Low	2	3	High
Low	-0.17	0.44	0.01	-0.30	Low	-0.63	1.75	0.02	-0.78
2	-0.07	0.34	0.20	-0.35	2	-0.46	2.33	1.10	-1.75
3	0.53	0.18	0.57	0.02	3	2.60	0.86	2.51	0.08
High	-0.16	0.00	0.34	-0.01	High	-0.72	-0.02	2.13	-0.15
Estimates for b					$t(b)$				
Low	0.93	0.94	1.08	0.95	Low	14.45	15.28	15.39	10.43
2	1.02	0.96	1.01	0.99	2	28.67	27.43	23.39	20.42
3	0.91	1.01	0.99	1.06	3	18.62	20.47	18.23	22.09
High	1.03	0.92	0.99	1.00	High	19.67	22.23	25.57	72.59
Estimates for s					$t(s)$				
Low	0.96	0.90	1.18	0.99	Low	9.82	9.66	11.01	7.15
2	1.06	0.93	0.99	1.17	2	19.72	17.52	14.95	15.95
3	0.55	0.62	0.55	0.71	3	7.45	8.26	6.60	9.68
High	-0.01	0.03	0.04	-0.07	High	-0.16	0.05	0.68	-3.17
Estimates for h					$t(h)$				
Low	0.32	0.36	0.24	0.13	Low	3.23	3.77	2.19	0.94
2	-0.14	0.07	0.19	0.18	2	-2.52	1.36	2.86	2.35
3	0.01	0.03	-0.03	-0.02	3	0.15	0.33	-0.31	-0.28
High	0.24	0.01	-0.20	0.06	High	2.91	0.09	-3.30	2.60
Estimates for r					$t(r)$				
Low	-0.39	-0.35	-0.24	0.11	Low	-3.89	-3.67	-2.15	0.80
2	-0.47	-0.14	-0.15	0.43	2	-8.44	-2.63	-2.27	5.67
3	-0.48	-0.26	-0.27	-0.23	3	-6.28	-3.30	-3.32	-3.11
High	-1.15	-0.30	-0.18	0.22	High	-14.08	-4.58	-2.91	10.31
Estimates for c					$t(c)$				
Low	0.09	-0.03	-0.14	0.02	Low	0.78	-0.29	-1.13	0.13
2	0.00	0.07	-0.05	-0.01	2	-0.01	1.08	-0.70	-0.07
3	0.01	-0.07	-0.05	0.03	3	0.07	-0.77	-0.57	0.31
High	0.02	0.04	0.31	-0.12	High	0.25	0.53	4.56	-4.90
Adjusted R^2									
Low	0.59	0.62	0.60	0.39					
2	0.84	0.82	0.77	0.74					
3	0.69	0.73	0.68	0.75					
High	0.81	0.81	0.84	0.98					

Table 6: Results for the Fama-French five-factor model for Size-Investment double sorted portfolios.

Regression estimates (left panel) for the different portfolios are based on equation (4) together with the corresponding t-statistics (right panel).

Estimates for α					$t(\alpha)$				
Size/Inv	Low	2	3	High	Size/Inv	Low	2	3	High
Low	-0.14	0.17	1.18	0.64	Low	-0.56	0.63	3.07	-1.89
2	-0.10	0.26	0.44	-0.21	2	-0.57	1.53	2.59	-1.33
3	0.12	0.38	0.48	0.34	3	0.46	2.31	2.66	1.84
High	-0.11	-0.01	0.16	-0.06	High	-0.50	-0.04	0.90	-0.40
Estimates for β					$t(\beta)$				
Low	1.00	0.92	0.82	1.10	Low	16.66	13.90	8.89	13.50
2	1.19	0.77	0.97	1.04	2	26.97	18.66	23.84	27.25
3	1.03	0.96	0.89	1.09	3	16.25	24.07	20.60	24.14
High	0.98	1.07	1.02	1.01	High	19.09	29.32	24.42	25.92
Estimates for s					$t(s)$				
Low	1.02	0.85	1.01	1.01	Low	11.14	8.46	7.19	8.21
2	1.28	0.75	0.91	1.16	2	19.12	11.95	14.78	20.04
3	0.76	0.58	0.46	0.67	3	7.87	9.54	7.00	9.79
High	0.13	0.01	-0.17	0.07	High	1.72	0.23	-2.74	1.20
Estimates for h					$t(h)$				
Low	0.40	0.23	0.23	0.03	Low	4.33	2.29	1.61	0.26
2	-0.07	0.22	0.03	0.14	2	-1.09	3.43	0.43	2.29
3	0.01	0.08	-0.03	-0.01	3	0.14	1.26	-0.45	-0.16
High	-0.03	0.14	-0.07	-0.04	High	-0.44	2.42	-1.05	-0.69
Estimates for r					$t(r)$				
Low	-0.28	-0.26	-0.47	-0.19	Low	-3.01	-2.53	-3.24	-1.51
2	-0.06	-0.11	-0.25	-0.12	2	-0.83	-1.68	-3.91	-1.93
3	-0.22	-0.26	-0.34	-0.33	3	-2.23	-4.18	-5.03	-4.71
High	-0.21	-0.23	0.20	-0.13	High	-2.57	-4.06	3.02	-2.13
Estimates for c					$t(c)$				
Low	0.21	0.10	-0.37	-0.34	Low	2.03	0.86	-2.29	-2.37
2	0.47	0.09	-0.16	-0.37	2	6.10	1.17	-2.23	-5.46
3	0.25	-0.02	-0.16	-0.12	3	2.21	-0.25	-2.13	-1.45
High	0.59	0.06	0.07	-0.56	High	6.57	0.89	0.91	-8.11
Adjusted R^2									
Low	0.66	0.55	0.37	0.53					
2	0.82	0.69	0.78	0.84					
3	0.61	0.79	0.74	0.79					
High	0.73	0.89	0.87	0.88					

5 Conclusion

We implement the Fama-French five-factor model for the German market. Looking at the factors standalone, the newly introduced profitability and investment factor do not show attractive, statistically significant returns on a long/short basis for the German equity market. From a correlation point of view, the profitability factor is negatively correlated to the size factor, while the investment factor is positively correlated to the size and value factor. The findings in this paper related to the observation of an inverse size factor are coherent with other German studies, however it is opposing international research. While the SMB factor is the only factor with a negative sign in terms of average returns, this result is not statistically significant. However, the comparison of the SMB factor returns to other studies on the German market lead to the conclusion, that small sized companies did particularly bad during the 1990s and 2000s. Comparing the results of the three-factor-model to the results of the five-factor-model for the German equity market, the improvement in explainability is rather marginal. The average adjusted R^2 of the five-factor model increases only slightly compared to the three-factor model. The different kind of sorts for the left-hand side portfolios help to analyze the potential tautologic relationship of left-hand side and right-hand side portfolios. Generally speaking, many of the regression intercepts for the RMW and CMA factors show strong statistical significance, even if the double sort of the left-hand side portfolios have been performed on other variables. Considering the GRS test statistics, we find evidence that the Fama-French five-factor model captures the cross-sectional return variation when considering Size/BM sorted portfolios and to a lesser extent for Size/Op sorted portfolios. Overall, the international evidence with respect to the significance of the newly introduced factors, operating profitability and investments, in explaining asset returns, is not supported by the our implementation of the Fama-French five-factor model to the German stock market.

References

- ARTMANN, S., P. FINTER, A. KEMPF, S. KOCH, AND E. THEISSEN (2012): “The Cross-Section of German Stock Returns: New Data and New Evidence,” *Schmalenbach business review : Sbr*, 64, 20–43.
- BANZ, R. W. (1981): “The Relationship between Return and Market Value of Common Stocks,” *Journal of Financial Economics*, 9(1), 3–18.
- BRUECKNER, R., P. LEHMANN, M. SCHMIDT, AND R. STEHLE (2015): “Non-U.S. Multi-Factor Data Sets Should be Used with Caution,” *Working Paper, Humboldt University Berlin*.
- CARHART, M. (1997): “On Persistence in Mutual Fund Performance,” *Journal of Finance*, 52(1), 57–82.
- CHIAH, M., D. CHAI, A. ZHONG, AND S. LI (2016): “A Better Model? An Empirical Investigation of the Fama-French Five-factor Model in Australia,” *International Review of Finance*, 16, 595–638.
- DANIEL, K., S. TITMAN, AND K. C. J. WEI (2001): “Explaining the Cross-Section of Stock Returns in Japan: Factors or Characteristics?,” *The Journal of Finance*, 56(2), 743–766.

- FAMA, E. F., AND K. R. FRENCH (1993): “Common Risk Factors in the Returns on Stocks and Bonds,” *Journal of Financial Economics*, 33, 3–56.
- FAMA, E. F., AND K. R. FRENCH (2012): “Size, Value, and Momentum in International Stock Returns,” *Journal of Financial Economics*, 105, 457–472.
- FAMA, E. F., AND K. R. FRENCH (2015): “A Five-Factor Asset Pricing Model,” *Journal of Financial Economics*, 116, 1–22.
- FAMA, E. F., AND K. R. FRENCH (2017): “International Tests of a Five-Factor Asset Pricing Model,” *Journal of Financial Economics*, 123, 441–463.
- GIBBONS, R., T. ROSS, AND J. SHANKEN (1989): “A Test of the Efficiency of a Given Portfolio,” *Econometrica*, 57(2), 1121–1152.
- GLASER, M., AND M. WEBER (2003): “Momentum and Turnover: Evidence from the German Stock Market,” *Schmalenbach Business Review*, 55(2), 108–135.
- GRIFFIN, J. M. (2002): “Are the Fama and French Factors Global or Country Specific?,” *The Review of Financial Studies*, 15(3), 783–803.
- HANAUER, M. X., C. KASERER, AND M. S. RAPP (2011): “Risikofaktoren und Multifaktormodelle fuer den Deutschen Aktienmarkt (Risk Factors and Multi-Factor Models for the German Stock Market),” *CEFS Working Paper Series, No. 2011-01*.

- HARVEY, C. R., Y. LIU, AND H. ZHU (2016): "... and the Cross-Section of Expected Returns," *The Review of Financial Studies*, 29(1), 5–68.
- KUBOTA, K., AND H. TAKEHARA (2018): "Does the Fama and French Five-Factor Model Work Well in Japan?," *International Review of Finance*, 18, 137–146.
- L'HER, J.-F., T. MASMOUDI, AND J.-M. SURET (2004): "Evidence to support the four-factor pricing model from the Canadian stock market," *Journal of International Financial Markets, Institutions and Money*, 14(4), 313 – 328.
- MILLER, M., AND F. MODIGLIANI (1961): "Dividend Policy, Growth, and the Valuation of Shares," *The Journal of Business*, 34.
- SCHMIDT, P. S., U. VON ARX, A. SCHRIMPF, A. F. WAGNER, AND A. ZIEGLER (2017): "On the Construction of Common Size, Value and Momentum Factors in International Stock Markets: A Guide with Applications," *Swiss Finance Institute Research Paper 10-58*.
- SCHRIMPF, A., M. SCHROEDER, AND R. STEHLE (2007): "Cross-sectional Tests of Conditional Asset Pricing Models: Evidence from the German Stock Market," *European Financial Management*, 13, 880–907.
- ZIEGLER, A., M. SCHROEDER, A. SCHULZ, AND R. STEHLE (2007): "Multifaktormodelle zur Erklärung deutscher Aktienrenditen: Eine empirische Analyse," *Schmalenbachs Zeitschrift fuer betriebswirtschaftliche Forschung*, 59, 355–389.